



Alaska State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Alaska's Mathematics Standards.

We are committed to ensuring that our high-quality instructional materials align to Alaska's Mathematics Standards to fully support Alaska's students and teachers. The tables on the following pages, organized by Alaska Mathematics Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		Lessons
Counting & Cardinality		
K.CC.1	Count to 100 by ones and by tens.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.CC.2	Count forward beginning from a given number within the known sequence.	Mission 1, Topic G Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.CC.3	Write numbers from 0 to 20. Represent a number of objects with a written numeral 0 - 20 (with 0 representing a count of no objects).	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.CC.4	Understand the relationship between numbers and quantities; connect counting to cardinality. a. When counting objects, say the number names in standard order, pairing each object with one and only one number name and each number name with one and only one object. b. Understand that the last number name said tells the number of objects counted. The number of objects is the same regardless of their arrangement or the order in which they were counted. c. Understand that each successive number name refers to a quantity that is one larger.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G Mission 1, Topic H Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.CC.5	Count to answer "how many?" questions about as many as 20 things arranged in a line, a rectangular array or a circle, or as many as 10 things in a scattered configuration; given a number from 1-20, count out that many objects.	Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic A

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
K.CC.6	Identify whether the number of objects in one group is greater than, less than, or equal to the number of objects in another group (e.g., by using matching, counting, or estimating strategies).	Mission 3, Topic B Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H Mission 5, Topic E
K.CC.7	Compare and order two numbers between 1 and 10 presented as written numerals.	Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H
Operations & Algebraic Thinking		
K.OA.1	Represent addition and subtraction with objects, fingers, mental images, drawings, sounds (e.g., claps) acting out situations, verbal explanations, expressions, or equations.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
K.OA.2	Add or subtract whole numbers to 10 (e.g., by using objects or drawings to solve word problems).	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic F Mission 4, Topic G Mission 4, Topic H
K.OA.3	Decompose numbers less than or equal to 10 into pairs in more than one way (e.g., by using objects or drawings, and record each decomposition by a drawing or equation).	Mission 1, Topic C Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic G
K.OA.4	For any number from 1 - 4, find the number that makes 5 when added to the given number and, for any number from 1 - 9, find the number that makes 10 when added to the given number (e.g., by using	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic H

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		Lessons
	objects, drawings or 10 frames) and record the answer with a drawing or equation.	
K.OA.5	Fluently add and subtract numbers up to 5.	Mission 4, Topic A
K.OA.6	Recognize, identify and continue simple patterns of color, shape, and size.	Mission 1, Topic H Mission 4, Topic B Mission 4, Topic H Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
Numbers & Operations in Base Ten		
K.NBT.1	Compose and decompose numbers from 11 to 19 into ten ones and some further ones (e.g., by using objects or drawings) and record each composition and decomposition by a drawing or equation (e.g., $18 = 10 + 8$); understand that these numbers are composed of ten ones and one, two, three, four, five, six, seven, eight or nine ones.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
Measurement & Data		
K.MD.1	Describe measurable attributes of objects (e.g., length or weight). Match measuring tools to attribute (e.g., ruler to length). Describe several measurable attributes of a single object.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
K.MD.2	Make comparisons between two objects with a measurable attribute in common, to see which object has “more of”/“less of” the attribute, and describe the difference.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F Mission 3, Topic H
K.MD.3	Classify objects into given categories (attributes). Count the number of objects in each category (limit category counts to be less than or equal to 10).	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 2, Topic C
K.MD.4	Name in sequence the days of the week.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
K.MD.5	Tell time to the hour using both analog and digital clocks.	Grade 1, Mission 5, Topic D
K.MD.6	Identify coins by name.	Grade 1, Mission 6, Topic E
Geometry		
K.G.1	Describe objects in the environment using names of shapes and describe their relative positions (e.g., above, below, beside, in front of, behind, next to).	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic B
K.G.2	Name shapes regardless of their orientation or overall size.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic B
K.G.3	Identify shapes as twodimensional (flat) or three-dimensional (solid).	Mission 2, Topic C
K.G.4	Analyze and compare two- and three-dimensional shapes, in different sizes and orientations, using informal language to describe their similarities, differences, parts (e.g., number of sides and vertices), and other attributes (e.g., having sides of equal lengths).	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A Mission 6, Topic B
K.G.5	Build shapes (e.g., using sticks and clay) and draw shapes.	Mission 6, Topic A
K.G.6	Put together two-dimensional shapes to form larger shapes (e.g., join two triangles with full sides touching to make a rectangle).	Mission 6, Topic B

1st Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
Counting & Cardinality		
1.CC.1	Skip count by 2s and 5s.	Grade 2, Mission 6, Topic D Grade 2, Mission 8, Topic D
1.CC.2	Use ordinal numbers correctly when identifying object position (e.g., first, second, third, etc.).	Kindergarten, Mission 6, Topic A
1.CC.3	Order numbers from 1-100. Demonstrate ability in counting forward and backward.	Mission 4, Topic A Mission 6, Topic B
1.CC.4	Count a large quantity of objects by grouping into 10s and counting by 10s and 1s to find the quantity.	Mission 2, Topic D Mission 4, Topic A
1.CC.5	Use the symbols for greater than, less than or equal to when comparing two numbers or groups of objects.	Mission 4, Topic B Mission 6, Topic B
1.CC.6	Estimate how many and how much in a given set to 20 and then verify estimate by counting	<i>Zearn Math includes digital activities that require counting a set of objects; however, the Zearn Math digital activities do not require students to estimate how many and how much is in a given set of objects. It is recommended that teachers use supplemental materials to fully address this standard.</i>
Operations & Algebraic Thinking		
1.OA.1	Use addition and subtraction strategies to solve word problems (using numbers up to 20), involving situations of adding to, taking from, putting together, taking apart and comparing, with unknowns in all positions, using a number line (e.g., by using objects, drawings and equations). Record and explain using equation symbols and a symbol for the unknown number to represent the problem.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic G Mission 1, Topic H Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
		Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 6, Topic A Mission 6, Topic F
1.OA.2	Solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20 (e.g., by using objects, drawings and equations). Record and explain using equation symbols and a symbol for the unknown number to represent the problem.	Mission 2, Topic A
1.OA.3	Apply properties of operations as strategies to add and subtract. (Students need not know the name of the property.)	Mission 1, Topic E Mission 1, Topic F Mission 2, Topic A Mission 2, Topic B
1.OA.4	Understand subtraction as an unknown-addend problem.	Mission 1, Topic G Mission 1, Topic H Mission 1, Topic I Mission 2, Topic B Mission 2, Topic C
1.OA.5	Relate counting to addition and subtraction (e.g., by counting on 2 to add 2).	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic G Mission 1, Topic H Mission 1, Topic I Mission 2, Topic B Mission 2, Topic C
1.OA.6	Add and subtract using numbers up to 20, demonstrating fluency for addition and subtraction up to 10. Use strategies such as: counting on, making ten ($8 + 6 = 8 + 2 + 4 = 10 + 4 = 14$), decomposing a number leading to a ten ($13 - 4 = 13 - 3 - 1 = 10 - 1 = 9$), using the relationship between addition and subtraction, such as fact families, ($8 + 4 = 12$ and $12 - 8 = 4$), creating equivalent but easier or known sums (e.g., adding $6 + 7$ by creating the known equivalent $6 + 6 + 1 = 12 + 1 = 13$).	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic F Mission 1, Topic I Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
1.OA.7	Understand the meaning of the equal sign (e.g., read equal sign as “same as”) and determine if equations involving addition and subtraction are true or false.	Mission 1, Topic E Mission 2, Topic B Mission 2, Topic C
1.OA.8	Determine the unknown whole number in an addition or subtraction equation.	Mission 1, Topic D Mission 1, Topic H Mission 2, Topic C
1.OA.9	Identify, continue and label patterns (e.g., aabb, abab). Create patterns using number, shape, size, rhythm or color	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic F Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 4, Topic D Mission 6, Topic B
Numbers & Operations in Base Ten		
1.NBT.1	Count to 120. In this range, read, write and order numerals and represent a number of objects with a written numeral.	Mission 4, Topic A Mission 6, Topic B
1.NBT.2	Model and identify place value positions of two digit numbers. Include: a. 10 can be thought of as a bundle of ten ones, called a "ten". b. The numbers from 11 to 19 are composed of a ten and one, two, three, four, five, six, seven, eight or nine ones. c. The numbers 10, 20, 30, 40, 50, 60, 70, 80, 90, refer to one, two, three, four, five, six, seven, eight or nine tens (and 0 ones).	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic B Mission 4, Topic F
1.NBT.3	Compare two two-digit numbers based on meanings of the tens and ones digits, recording the results of comparisons with the symbols $>$, $=$, $<$	Mission 4, Topic B Mission 6, Topic B
1.NBT.4	Add using numbers up to 100 including adding a two-digit number and a one-digit number and adding a two-digit number and a multiple of 10. Use: concrete models or drawings and strategies based on place value, properties of operations, and/or relationship between addition and subtraction. Relate the strategy to a written method and explain the reasoning used. Demonstrate in adding two-digit numbers, tens and	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic F Mission 6, Topic C Mission 6, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
	tens are added, ones and ones are added and sometimes it is necessary to compose a ten from ten ones.	
1.NBT.5	Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	Mission 2, Topic D Mission 4, Topic A Mission 6, Topic B
1.NBT.6	Subtract multiples of 10 up to 100. Use: concrete models or drawings, strategies based on place value, properties of operations, and/or the relationship between addition and subtraction. Relate the strategy to a written method and explain the reasoning used.	Mission 4, Topic C Mission 6, Topic C
Measurement & Data		
1.MD.1	Measure and compare three objects using standard or non-standard units.	Mission 3, Topic A Mission 3, Topic B
1.MD.2	Express the length of an object as a whole number of length units, by laying multiple copies of a shorter object (the length unit) end to end; understand that the length measurement of an object is the number of same-size length units that span it with no gaps or overlaps.	Mission 3, Topic B Mission 3, Topic C
1.MD.3	Tell and write time in half hours using both analog and digital clocks.	Mission 5, Topic D
1.MD.4	Read a calendar distinguishing yesterday, today and tomorrow. Read and write a date.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.MD.5	Recognize and read money symbols including \$ and ¢.	Grade 2, Mission 7, Topic B
1.MD.6	Identify values of coins (e.g., nickel = 5 cents, quarter = 25 cents). Identify equivalent values of coins up to \$1 (e.g., 5 pennies = 1 nickel, 5 nickels = 1 quarter).	Mission 4, Topic A Mission 6, Topic E
1.MD.7	Organize, represent and interpret data with up to three categories. Ask and answer comparison and quantity questions about the data.	Mission 3, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
Geometry		
1.G.1	Distinguish between defining attributes (e.g., triangles are closed and three-sided) versus non-defining attributes. Identify shapes that have nondefining attributes (e.g., color, orientation, overall size). Build and draw shapes given specified attributes.	Mission 5, Topic A
1.G.2	Compose (put together) twodimensional or three-dimensional shapes to create a larger, composite shape, and compose new shapes from the composite shape.	Mission 5, Topic B
1.G.3	Partition circles and rectangles into two and four equal shares. Describe the shares using the words, halves, fourths, and quarters and phrases half of, fourth of and quarter of. Describe the whole as two of or four of the shares. Understand for these examples that decomposing (break apart) into more equal shares creates smaller shares.	Mission 5, Topic C Mission 5, Topic D

2nd Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
Operations & Algebraic Thinking		
2.OA.1	Use addition and subtraction strategies to estimate, then solve one- and two-step word problems (using numbers up to 100) involving situations of adding to, taking from, putting together, taking apart and comparing, with unknowns in all positions (e.g., by using objects, drawings and equations). Record and explain using equation symbols and a symbol for the unknown number to represent the problem.	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F
2.OA.2	Fluently add and subtract using numbers up to 20 using mental strategies. Know from memory all sums of two onedigit numbers.	Mission 1, Topic A Mission 1, Topic B
2.OA.3	Determine whether a group of objects (up to 20) is odd or even (e.g., by pairing objects and comparing, counting by 2s). Model an even number as two equal groups of objects and then write an equation as a sum of two equal addends.	Mission 6, Topic D
2.OA.4	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns. Write an equation to express the total as repeated addition (e.g., array of 4 by 5 would be $5 + 5 + 5 = 20$).	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
2.OA.5	Identify, continue and label number patterns (e.g., aabb, abab). Describe a rule that determines and continues a sequence or pattern.	Mission 1, Topic B Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 5, Topic A Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
		Mission 6, Topic D Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
Numbers & Operations in Base Ten		
2.NBT.1	Model and identify place value positions of three digit numbers. Include: a. 100 can be thought of as a bundle of ten tens --called a "hundred". b. The numbers 100, 200, 300, 400, 500, 600, 700, 800, 900 refer to one, two, three, four, five, six, seven, eight, or nine hundreds (and 0 tens and 0 ones).	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NBT.2	Count up to 1000, skip-count by 5s, 10s and 100s.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.NBT.3	Read, write, order up to 1000 using base-ten numerals, number names and expanded form	Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NBT.4	Compare two three-digit numbers based on the meanings of the hundreds, tens and ones digits, using $>$, $=$, $<$ symbols to record the results.	Mission 3, Topic E Mission 3, Topic F
2.NBT.5	Fluently add and subtract using numbers up to 100. Use: strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 1, Topic B Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.NBT.6	Add up to four two-digit numbers using strategies based on place value and properties of operations.	Mission 4, Topic D Mission 6, Topic A

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
		Mission 7, Topic B Mission 8, Topic D
2.NBT.7	Add and subtract using numbers up to 1000. Use: concrete models or drawings and strategies based on place value, properties of operations, and/or relationship between addition and subtraction. Relate the strategy to a written method and explain the reasoning used. Demonstrate in adding or subtracting three-digit numbers, hundreds and hundreds are added or subtracted, tens and tens are added or subtracted, ones and ones are added or subtracted and sometimes it is necessary to compose a ten from ten ones or a hundred from ten tens.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D
2.NBT.8	Mentally add 10 or 100 to a given number 100-900 and mentally subtract 10 or 100 from a given number.	Mission 3, Topic G Mission 5, Topic A Mission 5, Topic D
2.NBT.9	Explain or illustrate the processes of addition or subtraction and their relationship using place value and the properties of operations	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D
Measurement & Data		
2.MD.1	Measure the length of an object by selecting and using standard tools such as rulers, yardsticks, meter sticks, and measuring tapes.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 7, Topic C Mission 7, Topic D Mission 7, Topic F Mission 8, Topic A
2.MD.2	Measure the length of an object twice using different length units for the two measurements. Describe how	Mission 2, Topic C Mission 7, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
	the two measurements relate to the size of the unit chosen.	
2.MD.3	Estimate, measure and draw lengths using whole units of inches, feet, yards, centimeters and meters.	Mission 2, Topic B Mission 2, Topic D Mission 7, Topic D
2.MD.4	Measure to compare lengths of two objects, expressing the difference in terms of a standard length unit.	Mission 2, Topic C Mission 2, Topic D Mission 7, Topic D
2.MD.5	Solve addition and subtraction word problems using numbers up to 100 involving length that are given in the same units (e.g., by using drawings of rulers). Write an equation with a symbol for the unknown to represent the problem.	Mission 2, Topic D Mission 7, Topic E Mission 7, Topic F
2.MD.6	Represent whole numbers as lengths from 0 on a number line diagram with equally spaced points corresponding to the numbers 0, 1, 2, ..., and represent whole-number sums and differences within 100 on a number line diagram.	Mission 7, Topic A Mission 7, Topic E Mission 7, Topic F
2.MD.7	Tell and write time to the nearest five minutes using a.m. and p.m. from analog and digital clocks.	Mission 8, Topic D
2.MD.8	Solve word problems involving dollar bills and coins using the \$ and ¢ symbols appropriately.	Mission 3, Topic D Mission 7, Topic B
2.MD.9	Collect, record, interpret, represent, and describe data in a table, graph or line plot.	Mission 7, Topic F
2.MD.10	Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart and compare problems using information presented in a bar graph.	Mission 7, Topic A
Geometry		
2.G.1	Identify and draw shapes having specified attributes, such as a given number of angles or a given number of	Mission 8, Topic A Mission 8, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
	equal faces compared visually, not by measuring. Identify triangles, quadrilaterals, pentagons, hexagons and cubes.	Mission 8, Topic C
2.G.2	Partition a rectangle into rows and columns of same-size squares and count to find the total number of them.	Mission 6, Topic C
2.G.3	Partition circles and rectangles into shares, describe the shares using the words <i>halves</i> , <i>thirds</i> , <i>half of</i> , <i>a third of</i> , etc., and describe the whole as two halves, three thirds, four fourths. Recognize that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D

3rd Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
Operations & Algebraic Thinking		
3.OA.1	Interpret products of whole numbers (e.g., interpret 5×7 as the total number of objects in 5 groups of 7 objects each).	Mission 1, Topic A Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
3.OA.2	Interpret whole-number quotients of whole numbers (e.g., interpret $56 \div 8$ as the number of objects in each share when 56 objects are partitioned equally into 8 shares, or as a number of shares when 56 objects are partitioned into equal shares of 8 objects each).	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.3	Use multiplication and division numbers up to 100 to solve word problems in situations involving equal groups, arrays, and measurement quantities (e.g., by using drawings and equations with a symbol for the unknown number to represent the problem).	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.4	Determine the unknown whole number in a multiplication or division equation relating three whole numbers.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.5	Make, test, support, draw conclusions and justify conjectures about properties of operations as strategies to multiply and divide. (Students need not use formal terms for these properties.) • Commutative property of multiplication: If $6 \times 4 = 24$ is known, then $4 \times 6 = 24$ is also known. • Associative property of multiplication: $3 \times 5 \times 2$ can be found by $3 \times 5 = 15$, then $15 \times 2 = 30$, or by $5 \times 2 = 10$, then $3 \times 10 = 30$. • Distributive property: Knowing that $8 \times 5 = 40$ and $8 \times 2 = 16$, one can find 8×7 as $8 \times (5 + 2) = (8 \times 5) + (8 \times 2) = 40 + 16 = 56$. • Inverse property (relationship) of multiplication and division.	Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F
3.OA.6	Understand division as an unknown-factor problem.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.7	Fluently multiply and divide numbers up to 100, using strategies such as the relationship between multiplication and division (e.g., knowing that $8 \times 5 = 40$, one knows $40 \div 5 = 8$) or properties of operations. By the end of Grade 3, know from memory all products of two onedigit numbers.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.8	Solve and create two-step word problems using any of the four operations. Represent these problems using equations with a symbol (box, circle, question mark) standing for the unknown quantity. Assess the	Mission 1, Topic F Mission 3, Topic C Mission 3, Topic E Mission 3, Topic F Mission 7, Topic A

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
	reasonableness of answers using mental computation and estimation strategies including rounding.	
3.OA.9	Identify arithmetic patterns (including patterns in the addition table or multiplication table) and explain them using properties of operations.	Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
Numbers & Operations in Base Ten		
3.NBT.1	Use place value understanding to round whole numbers to the nearest 10 or 100.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.NBT.2	Use strategies and/or algorithms to fluently add and subtract with numbers up to 1000, demonstrating understanding of place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E
3.NBT.3	Multiply one-digit whole numbers by multiples of 10 in the range 10-90 (e.g., 9×80 , 10×60) using strategies based on place value and properties of operations.	Mission 3, Topic F
Numbers & Operations - Fractions		
3.NF.1	Understand a fraction $1/b$ (e.g., $1/4$) as the quantity formed by 1 part when a whole is partitioned into b (e.g., 4) equal parts; understand a fraction a/b (e.g., $2/4$) as the quantity formed by a (e.g., 2) parts of size $1/b$. (e.g., $1/4$)	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
3.NF.2	Understand a fraction as a number on the number line; represent fractions on a number line diagram. a. Represent a fraction $1/b$ (e.g., $1/4$) on a number line diagram by defining the interval from 0 to 1 as the whole and partitioning it into b (e.g., 4) equal parts. Recognize that each part has size $1/b$ (e.g., $1/4$) and that the endpoint of the part based at 0 locates the number $1/b$ (e.g., $1/4$) on the number line. b. Represent a fraction a/b (e.g., $2/8$) on a number line diagram or ruler by marking off a lengths $1/b$ (e.g., $1/8$)	Mission 5, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
	from 0. Recognize that the resulting interval has size a/b (e.g., $2/8$) and that its endpoint locates the number a/b (e.g., $2/8$) on the number line.	
3.NF.3	Explain equivalence of fractions in special cases, and compare fractions by reasoning about their size. a. Understand two fractions as equivalent if they are the same size (modeled) or the same point on a number line. b. Recognize and generate simple equivalent fractions (e.g., $1/2 = 2/4$, $4/6 = 2/3$). Explain why the fractions are equivalent (e.g., by using a visual fraction model). c. Express and model whole numbers as fractions, and recognize and construct fractions that are equivalent to whole numbers.. d. Compare two fractions with the same numerator or the same denominator by reasoning about their size. Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions (e.g., by using a visual fraction model).	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Measurement & Data		
3.MD.1	Tell and write time to the nearest minute and measure time intervals in minutes. Solve word problems involving addition and subtraction of time intervals in minutes or hours (e.g., by representing the problem on a number line diagram or clock).	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.MD.2	Estimate and measure liquid volumes and masses of objects using standard units of grams (g), kilograms (kg), and liters (l). (Excludes compound units such as cm^3 and finding the geometric volume of a container.) Add, subtract, multiply, or divide to solve and create one-step word problems involving masses or volumes that are given in the same units (e.g., by using drawings, such as a beaker with a measurement scale, to represent the problem). (Excludes multiplicative comparison problems [problems involving notions of “times as much.”])	Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
3.MD.3	Select an appropriate unit of English, metric, or non-standard measurement to estimate the length, time, weight, or temperature.	Grade 2, Mission 2, Topic B Grade 2, Mission 2, Topic D Grade 2, Mission 7, Topic D Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.MD.4	Draw a scaled picture graph and a scaled bar graph to represent a data set with several categories. Solve one- and two-step “how many more” and “how many less” problems using information presented in scaled bar graphs.	Mission 6, Topic A
3.MD.5	Measure and record lengths using rulers marked with halves and fourths of an inch. Make a line plot with the data, where the horizontal scale is marked off in appropriate units—whole numbers, halves, or quarters.	Mission 6, Topic B Mission 7, Topic D
3.MD.6	Explain the classification of data from real-world problems shown in graphical representations. Use the terms minimum and maximum.	Grade 3, Mission 6, Topic B <i>This topic focuses on explaining the classification of data from real-world problems shown in graphical representations. It is recommended that teachers use supplemental materials to address the depth of this standard.</i>
3.MD.7	Recognize area as an attribute of plane figures and understand concepts of area measurement. a. A square with side length 1 unit is said to have “one square unit” and can be used to measure area. b. Demonstrate that a plane figure which can be covered without gaps or overlaps by n (e.g., 6) unit squares is said to have an area of n (e.g., 6) square units.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
3.MD.8	Measure areas by tiling with unit squares (square centimeters, square meters, square inches, square feet, and improvised units).	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C
3.MD.9	Relate area to the operations of multiplication and addition. a. Find the area of a rectangle with	Mission 4, Topic A Mission 4, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
	wholenumber side lengths by tiling it, and show that the area is the same as would be found by multiplying the side lengths. b. Multiply side lengths to find areas of rectangles with whole number side lengths in the context of solving real world and mathematical problems, and represent whole-number products as rectangular areas in mathematical reasoning. c. Use area models (rectangular arrays) to represent the distributive property in mathematical reasoning. Use tiling to show in a concrete case that the area of a rectangle with whole-number side lengths a and $b + c$ is the sum of $a \times b$ and $a \times c$. Recognize area as additive. Find areas of rectilinear figures by decomposing them into non-overlapping rectangles and adding the areas of the non-overlapping parts, applying this technique to solve real world problems.	Mission 4, Topic C Mission 4, Topic D
3.MD.10	Solve real world and mathematical problems involving perimeters of polygons, including: finding the perimeter given the side lengths, finding an unknown side length, exhibiting rectangles with the same perimeter and different areas, exhibiting rectangles with the same area and different perimeters.	Mission 7, Topic C Mission 7, Topic D Mission 7, Topic E
Geometry		
3.G.1	Categorize shapes by different attribute classifications and recognize that shared attributes can define a larger category. Generalize to create examples or non-examples.	Mission 7, Topic B
3.G.2	Partition shapes into parts with equal areas. Express the area of each part as a unit fraction of the whole	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C

4th Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
Operations & Algebraic Thinking		
4.OA.1	Interpret a multiplication equation as a comparison e.g., interpret $35 = 5 \times 7$ as a statement that 35 is 5 groups of 7 and 7 groups of 5 (Commutative property). Represent verbal statements of multiplicative comparisons as multiplication equations.	Mission 1, Topic A Mission 3, Topic A Mission 3, Topic B Mission 3, Topic D Mission 7, Topic A
4.OA.2	Multiply or divide to solve word problems involving multiplicative comparison (e.g., by using drawings and equations with a symbol for the unknown number to represent the problem or missing numbers in an array). Distinguish multiplicative comparison from additive comparison.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 5, Topic G Mission 7, Topic A Mission 7, Topic B
4.OA.3	Solve multistep word problems posed with whole numbers and having whole-number answers using the four operations, including problems in which remainders must be interpreted. Represent these problems using equations with a letter standing for the unknown quantity. Assess the reasonableness of answers using mental computation and estimation strategies including rounding.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 3, Topic H Mission 7, Topic B Mission 7, Topic C
4.OA.4	a. Find all factor pairs for a whole number in the range 1–100. b. Explain the correlation/differences between multiples and factors. c. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite.	Mission 3, Topic F
4.OA.5	Generate a number, shape pattern, table, t-chart, or input/output function that follows a given rule. Identify apparent features of the pattern that were not explicit in the rule itself. Be able to express the pattern in algebraic terms.	Mission 5, Topic H

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
4.OA.6	Extend patterns that use addition, subtraction, multiplication, division or symbols, up to 10 terms, represented by models (function machines), tables, sequences, or in problem situations.	Mission 5, Topic H
Numbers & Operations in Base Ten		
4.NBT.1	Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right.	Mission 1, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic G
4.NBT.2	Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form. Compare two multi-digit numbers based on the value of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NBT.3	Use place value understanding to round multi-digit whole numbers to any place using a variety of estimation methods; be able to describe, compare, and contrast solutions.	Mission 1, Topic C
4.NBT.4	Fluently add and subtract multidigit whole numbers using any algorithm. Verify the reasonableness of the results.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NBT.5	Multiply a whole number of up to four digits by a one-digit whole number, and multiply two two-digit numbers, using strategies based on place value and the properties of operations. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
4.NBT.6	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G
Numbers & Operations - Fractions		

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
4.NF.1	Explain why a fraction a/b is equivalent to a fraction $(n \times a)/(n \times b)$ by using visual fraction models, with attention to how the number and size of the parts differ even though the two fractions themselves are the same size. Use this principle to recognize and generate equivalent fractions.	Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 6, Topic B
4.NF.2	Compare two fractions with different numerators and different denominators (e.g., by creating common denominators or numerators, or by comparing to a benchmark fraction such as $1/2$). Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols $>$, $=$, or $<$, and justify the conclusions (e.g., by using a visual fraction model).	Mission 5, Topic C Mission 5, Topic E
4.NF.3	Understand a fraction a/b with $a > 1$ as a sum of fractions $1/b$. a. Understand addition and subtraction of fractions as joining and separating parts referring to the same whole. b. Decompose a fraction into a sum of fractions with the same denominator in more than one way, recording each decomposition by an equation. Justify decompositions (e.g., by using a visual fraction model). c. Add and subtract mixed numbers with like denominators (e.g., by replacing each mixed number with an equivalent fraction, and/or by using properties of operations and the relationship between addition and subtraction). d. Solve word problems involving addition and subtraction of fractions referring to the same whole and having like denominators (e.g., by using visual fraction models and equations to represent the problem).	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic D
4.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction by a whole number. a. Understand a fraction a/b as a multiple of $1/b$. b. Understand a multiple of a/b as a multiple of $1/b$, and use this understanding to multiply a fraction by a whole number. c. Solve word problems involving multiplication of a fraction by a whole number (e.g., by using visual fraction models and equations to represent the problem). Check for the reasonableness of the answer.	Mission 5, Topic A Mission 5, Topic E Mission 5, Topic G

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
4.NF.5	Express a fraction with denominator 10 as an equivalent fraction with denominator 100, and use this technique to add two fractions with respective denominators 10 and 100.	Mission 6, Topic B Mission 6, Topic D Mission 6, Topic E
4.NF.6	Use decimal notation for fractions with denominators 10 or 100.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic D Mission 6, Topic E
4.NF.7	Compare two decimals to hundredths by reasoning about their size. Recognize that comparisons are valid only when the two decimals refer to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions (e.g., by using a visual model).	Mission 6, Topic B Mission 6, Topic C
Measurement & Data		
4.MD.1	Know relative sizes of measurement units within one system of units including km, m, cm; kg, g; lb, oz.; l, ml; hr, min, sec. Within a single system of measurement, express measurements in a larger unit in terms of a smaller unit. Record measurement equivalents in a twocolumn table.	Mission 2, Topic A Mission 2, Topic B Mission 5, Topic D Mission 5, Topic F Mission 5, Topic G Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
4.MD.2	Use the four operations to solve word problems involving distances, intervals of time, liquid volumes, masses of objects, and money, including problems involving simple fractions or decimals, and problems that require expressing measurements given in a larger unit in terms of a smaller unit. Represent measurement quantities using diagrams such as number line diagrams that feature a measurement scale.	Mission 2, Topic A Mission 2, Topic B Mission 5, Topic D Mission 5, Topic F Mission 5, Topic G Mission 6, Topic C Mission 6, Topic E Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
4.MD.3	Apply the area and perimeter formulas for rectangles in real world and mathematical problems.	Mission 3, Topic A Mission 3, Topic H
4.MD.4	Solve real-world problems involving elapsed time between U.S. time zones (including Alaska Standard time).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.MD.5	Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$). Solve problems involving addition and subtraction of fractions by using information presented in line plots.	Mission 5, Topic E Mission 5, Topic G
4.MD.6	Explain the classification of data from real-world problems shown in graphical representations including the use of terms range and mode with a given set of data.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.MD.7	Recognize angles as geometric shapes that are formed wherever two rays share a common endpoint, and understand the following concepts of angle measurement: a. An angle is measured with reference to a circle with its center at the common endpoint of the rays, by considering the fraction of the circular arc between the points where the two rays intersect the circle. An angle that turns through $\frac{1}{360}$ of a circle is called a “one-degree angle,” and can be used to measure angles. b. An angle that turns through n onedegree angles is said to have an angle measure of n degrees.	Mission 4, Topic B
4.MD.8	Measure and draw angles in whole-number degrees using a protractor. Estimate and sketch angles of specified measure.	Mission 4, Topic B
4.MD.9	Recognize angle measure as additive. When an angle is divided into non-overlapping parts, the angle measure of the whole is the sum of the angle measures of the parts. Solve addition and subtraction problems to find unknown angles on a diagram in real world and mathematical problems (e.g., by using an equation with a symbol for the unknown angle measure).	Mission 4, Topic C

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
Geometry		
4.G.1	Draw points, lines, line segments, rays, angles (right, acute, obtuse), and perpendicular, parallel, and intersecting line segments. Identify these in twodimensional (plane) figures.	Mission 4, Topic A Mission 4, Topic D
4.G.2	Classify two-dimensional (plane) figures based on the presence or absence of parallel or perpendicular lines, or the presence or absence of angles of a specified size. Recognize right triangles as a category, and identify right triangles.	Mission 4, Topic D
4.G.3	Recognize a line of symmetry for a two-dimensional (plane) figure as a line across the figure such that the figure can be folded along the line into matching parts. Identify line-symmetric figures and draw lines of symmetry	Mission 4, Topic D

5th Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
Operations & Algebraic Thinking		
5.OA.1	Use parentheses to construct numerical expressions, and evaluate numerical expressions with these symbols.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
5.OA.2	Write simple expressions that record calculations with numbers, and interpret numerical expressions without evaluating them.	Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic H Mission 6, Topic B
5.OA.3	Generate two numerical patterns using two given rules. Identify apparent relationships between corresponding terms. Form ordered pairs consisting of corresponding terms from the two patterns, and graph the ordered pairs on a coordinate plane.	Mission 6, Topic B Mission 6, Topic D
Numbers & Operations in Base Ten		
5.NBT.1	Recognize that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and $\frac{1}{10}$ of what it represents in the place to its left.	Mission 1, Topic A Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
5.NBT.2	Explain and extend the patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain and extend the patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of 10. Use whole-number exponents to denote powers of 10.	Mission 1, Topic A Mission 1, Topic D Mission 1, Topic E Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E Mission 2, Topic G
5.NBT.3	Read, write, and compare decimals to thousandths. a. Read and write decimals to thousandths using	Mission 1, Topic B Mission 1, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
	base-ten numerals, number names, and expanded form [e.g., $347.392 = 3 \times 100 + 4 \times 10 + 7 \times 1 + 3 (1/10) + 9 (1/100) + 2 (1/1000)$]. b. Compare two decimals to thousandths place based on meanings of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 1, Topic E Mission 1, Topic F
5.NBT.4	Use place values understanding to round decimals to any place.	Mission 1, Topic C
5.NBT.5	Fluently multiply multi-digit whole numbers using a standard algorithm.	Mission 2, Topic B Mission 2, Topic D
5.NBT.6	Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, number lines, real life situations, and/or area models.	Mission 2, Topic E Mission 2, Topic F Mission 2, Topic H
5.NBT.7	Add, subtract, multiply, and divide decimals to hundredths, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between the operations. Relate the strategy to a written method and explain their reasoning in getting their answers.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 2, Topic C Mission 2, Topic D Mission 2, Topic G Mission 2, Topic H Mission 4, Topic E Mission 4, Topic G
Numbers & Operations - Fractions		
5.NF.1	Add and subtract fractions with unlike denominators (including mixed numbers) by replacing given fractions with equivalent fractions in such a way as to produce an equivalent sum or difference of fractions with like denominators.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D
5.NF.2	Solve word problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators (e.g., by using visual fraction models or equations to represent the	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
	problem). Use benchmark fractions and number sense of fractions to estimate mentally and check the reasonableness of answers.	
5.NF.3	Interpret a fraction as division of the numerator by the denominator ($a/b = a \div b$). Solve word problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers (e.g., by using visual fraction models or equations to represent the problem).	Mission 3, Topic A Mission 4, Topic B
5.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction or whole number by a fraction. a. Interpret the product $(a/b) \times q$ as a parts of a partition of q into b equal parts; equivalently, as the result of a sequence of operations $a \times q \div b$. b. Find the area of a rectangle with fractional side lengths by tiling it with unit squares of the appropriate unit fraction side lengths, and show that the area is the same as would be found by multiplying the side lengths. Multiply fractional side lengths to find areas of rectangles, and represent fraction products as rectangular areas.	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 5, Topic C
5.NF.5	Interpret multiplication as scaling (resizing), by: a. Comparing the size of a product to the size of one factor on the basis of the size of the other factor, without performing the indicated multiplication. b. Explaining why multiplying a given number by a fraction greater than 1 results in a product greater than the given number (recognizing multiplication by whole numbers greater than 1 as a familiar case); explaining why multiplying a given number by a fraction less than 1 results in a product smaller than the given number; and relating the principle of fraction equivalence $a/b = (n \times a)/(n \times b)$ to the effect of multiplying a/b by 1. (Division of a fraction by a fraction is not a requirement at this grade.)	Mission 4, Topic F
5.NF.6	Solve real world problems involving multiplication of fractions and mixed numbers (e.g., by using visual fraction models or equations to represent the problem).	Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 5, Topic C

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
5.NF.7	Apply and extend previous understandings of division to divide unit fractions by whole numbers and whole numbers by unit fractions. a. Interpret division of a unit fraction by a non-zero whole number, and compute such quotients. b. Interpret division of a whole number by a unit fraction, and compute such quotients. c. Solve real world problems involving division of unit fractions by non-zero whole numbers and division of whole numbers by unit fractions (e.g., by using visual fraction models and equations to represent the problem).	Mission 4, Topic G
Measurement & Data		
5.MD.1	Identify, estimate measure, and convert equivalent measures within systems English length (inches, feet, yards, miles) weight (ounces, pounds, tons) volume (fluid ounces, cups, pints, quarts, gallons) temperature (Fahrenheit) Metric length (millimeters, centimeters, meters, kilometers) volume (milliliters, liters), temperature (Celsius), (e.g., convert 5 cm to 0.05 m), and use these conversions in solving multi-step, real world problems using appropriate tools.	Mission 1, Topic A Mission 2, Topic D Mission 4, Topic C Mission 4, Topic E
5.MD.2	Solve real-world problems involving elapsed time between world time zones.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.MD.3	Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$). Solve problems involving information presented in line plots.	Mission 4, Topic A
5.MD.4	Explain the classification of data from real-world problems shown in graphical representations including the use of terms mean and median with a given set of data.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.MD.5	Recognize volume as an attribute of solid figures and understand concepts of volume measurement. a. A cube with side length 1 unit, called a “unit cube,” is said to have “one cubic unit” of volume, and can be used to measure volume. b. A solid figure that can be	Mission 5, Topic A Mission 5, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
	packed without gaps or overlaps using n unit cubes is said to have a volume of n cubic units.	
5.MD.6	Estimate and measure volumes by counting unit cubes, using cubic cm, cubic in, cubic ft, and non-standard units	Mission 5, Topic A
5.MD.5	Relate volume to the operations of multiplication and addition and solve real world and mathematical problems involving volume. a. Estimate and find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes, and show that the volume is the same as would be found by multiplying the edge lengths, equivalently by multiplying the height by the area of the base. Demonstrate the associative property of multiplication by using the product of three whole numbers to find volumes (length $\times$ width $\times$ height). b. Apply the formulas $V = l \times w \times h$ and $V = b \times h$ for rectangular prisms to find volumes of right rectangular prisms with whole number edge lengths in the context of solving real world and mathematical problems. c. Recognize volume as additive. Find volumes of solid figures composed of two non-overlapping right rectangular prisms by adding the volumes of the nonoverlapping parts, applying this technique to solve real world problems.	Mission 5, Topic B
Geometry		
5.G.1	Use a pair of perpendicular number lines, called axes, to define a coordinate system, with the intersection of the lines (the origin) arranged to coincide with the 0 on each line and a given point in the plane located by using an ordered pair of numbers, called its coordinates. Understand that the first number indicates how far to travel from the origin in the direction of one axis, and the second number indicates how far to travel in the direction of the second axis, with the convention that the names of the two axes and the coordinates correspond (e.g., x -axis and x -coordinate, y -axis and y -coordinate).	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
5.G.2	Represent real world and mathematical problems by graphing points in the first quadrant of the coordinate plane, and interpret coordinate values of points in the context of the situation.	Mission 6, Topic C Mission 6, Topic D
5.G.3	Understand that attributes belonging to a category of two-dimensional (plane) figures also belong to all subcategories of that category.	Mission 5, Topic D
5.G.4	Classify two-dimensional (plane) figures in a hierarchy based on attributes and properties.	Mission 5, Topic D

6th Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
Ratios & Proportional Relationships		
6.RP.1	Write and describe the relationship in real life context between two quantities using ratio language.	Mission 2, Topic A Mission 2, Topic B
6.RP.2	Understand the concept of a unit rate (a/b associated with a ratio $a:b$ with $b \neq 0$, and use rate language in the context of a ratio relationship) and apply it to solve real world problems (e.g., unit pricing, constant speed).	Mission 2, Topic C Mission 3, Topic A Mission 3, Topic C
6.RP.3	Use ratio and rate reasoning to solve real-world and mathematical problems (e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations). a. Make tables of equivalent ratios relating quantities with whole number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios, and understand equivalencies. b. Solve unit rate problems including those involving unit pricing and constant speed. c. Find a percent of a quantity as a rate per 100 (e.g., 30% of a quantity means $30/100$ times the quantity); solve problems involving finding the whole, given a part and the percent. d. Use ratio reasoning to convert measurement units between given measurement systems (e.g., convert kilometers to miles); manipulate and transform units appropriately when multiplying or dividing quantities.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Mission 2, Topic F Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F Mission 6, Topic B Mission 6, Topic D
The Number System		
6.NS.1	Interpret and compute quotients of fractions, and solve word problems involving division of fractions by fractions (e.g., by using visual fraction models and equations to represent the problem).	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
6.NS.2	Fluently multiply and divide multi-digit whole numbers using the standard algorithm. Express the remainder as a whole number, decimal, or simplified fraction; explain or justify your choice based on the context of the problem.	Mission 5, Topic D Grade 5, Mission 2, Topic B Grade 5, Mission 2, Topic D
6.NS.3	Fluently add, subtract, multiply, and divide multi-digit decimals using the standard algorithm for each operation. Express the remainder as a terminating decimal, or a repeating decimal, or rounded to a designated place value.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 6, Topic A Mission 8, Topic C
6.NS.4	Find the greatest common factor of two whole numbers less than or equal to 100 and the least common multiple of two whole numbers less than or equal to 12. Use the distributive property to express a sum of two whole numbers 1– 100 with a common factor as a multiple of a sum of two whole numbers with no common factor.	Mission 7, Topic D
6.NS.5	Understand that positive and negative numbers describe quantities having opposite directions or values (e.g., temperature above/below zero, elevation above/below sea level, credits/debits, positive/negative electric charge); use positive and negative numbers to represent quantities in real-world contexts, explain the meaning of 0 in each situation.	Mission 7, Topic A
6.NS.6	Understand a rational number as a point on the number line. Extend number line diagrams and coordinate axes familiar from previous grades to represent points on the line and in the plane with negative number coordinates. a. Recognize opposite signs of numbers as indicating locations on opposite sides of 0 on the number line; Recognize that the opposite of the opposite of a number is the number itself [e.g., $-(-3) = 3$] and that 0 is its own opposite. b. Understand signs of numbers in ordered pairs as indicating locations in quadrants of the coordinate plane; recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes. c. Find and	Mission 7, Topic A Mission 7, Topic C

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
	position integers and other rational numbers on a horizontal or vertical number line diagram; find and position pairs of integers and other rational numbers on a coordinate plane.	
6.NS.7	Understand ordering and absolute value of rational numbers. a. Interpret statements of inequality as statements about the relative position of two numbers on a number line diagram. b. Write, interpret, and explain statements of order for rational numbers in realworld contexts. c. Understand the absolute value of a rational number as its distance from 0 on the number line; interpret absolute value as magnitude for a positive or negative quantity in a real-world situation. d. Distinguish comparisons of absolute value from statements about order.	Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
6.NS.8	Solve real-world and mathematical problems by graphing points in all four quadrants of the coordinate plane. Include use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	Mission 7, Topic C
Expressions & Equations		
6.EE.1	Write and evaluate numerical expressions involving whole-number exponents.	Mission 1, Topic F Mission 6, Topic C
6.EE.2	Write, read, and evaluate expressions in which letters stand for numbers. a. Write expressions that record operations with numbers and with letters standing for numbers. b. Identify parts of an expression using mathematical terms (sum, term, product, factor, quotient, coefficient); view one or more parts of an expression as a single entity. c. Evaluate expressions and formulas. Include formulas used in real-world problems. Perform arithmetic operations, including those involving whole number exponents, in the conventional order with or without parentheses. (Order of Operations)	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic F Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.EE.3	Apply the properties of operations to generate equivalent expressions. Model (e.g., manipulatives, graph paper) and apply the distributive, commutative,	Mission 6, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
	identity, and inverse properties with integers and variables by writing equivalent expressions.	
6.EE.4	Identify when two expressions are equivalent (i.e., when the two expressions name the same number regardless of which value is substituted into them).	Mission 5, Topic D Mission 6, Topic B
6.EE.5	Understand solving an equation or inequality as a process of answering a question: which values from a specified set, if any, make the equation or inequality true? Use substitution to determine whether a given number in a specified set makes an equation or inequality true.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.EE.6	Use variables to represent numbers and write expressions when solving a real-world or mathematical problem; understand that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
6.EE.7	Solve real-world and mathematical problems by writing and solving equations of the form $x + p = q$ and $px = q$ for cases in which p , q and x are all nonnegative rational numbers.	Mission 6, Topic A Mission 6, Topic B
6.EE.8	Write an inequality of the form $x > c$ or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. Recognize that inequalities of the form $x > c$ or $x < c$ have infinitely many solutions; represent solutions of such inequalities on number line diagrams.	Mission 7, Topic B
6.EE.9	Use variables to represent two quantities in a real-world problem that change in relationship to one another; write an equation to express one quantity, thought of as the dependent variable, in terms of the other quantity, thought of as the independent variable. Analyze the relationship between the dependent and independent variables using graphs and tables, and relate these to the equation.	Mission 6, Topic D
Geometry		

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
6.G.1	Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing or decomposing into other polygons (e.g., rectangles and triangles). Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic G Mission 4, Topic D
6.G.2	Apply the standard formulas to find volumes of prisms. Use the attributes and properties (including shapes of bases) of prisms to identify, compare or describe three-dimensional figures including prisms and cylinders.	Mission 1, Topic E Mission 4, Topic D Mission 4, Topic E Grade 8, Mission 5, Topic D
6.G.3	Draw polygons in the coordinate plane given coordinates for the vertices; determine the length of a side joining the coordinates of vertices with the same first or the same second coordinate. Apply these techniques in the context of solving real-world and mathematical problems	Mission 7, Topic C
6.G.4	Represent three-dimensional figures (e.g., prisms) using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topics E Mission 1, Topics F Mission 1, Topics G
6.G.5	Identify, compare or describe attributes and properties of circles (radius, and diameter).	Grade 7, Mission 3, Topic A
Statistics & Probability		
6.SP.1	Recognize a statistical question as one that anticipates variability in the data related to the question and accounts for it in the answers.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic D
6.SP.2	Understand that a set of data has a distribution that can be described by its center (mean, median, or mode), spread (range), and overall shape and can be used to answer a statistical question.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic E
6.SP.3	Recognize that a measure of center (mean, median, or mode) for a numerical data set summarizes all of its values with a single number, while a measure of	Mission 8, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
	variation (range) describes how its values vary with a single number.	
6.SP.4	Display numerical data in plots on a number line, including dot or line plots, histograms and box (box and whisker) plots.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
6.SP.5	Summarize numerical data sets in relation to their context, such as by: a. Reporting the number of observations (occurrences). b. Describing the nature of the attribute under investigation, including how it was measured and its units of measurement. c. Giving quantitative measures of center (median and/or mean) and variability (interquartile range), as well as describing any overall pattern and any outliers with reference to the context in which the data were gathered. d. Relating the choice of measures of center and variability to the shape of the data distribution and the context in which the data were gathered.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
6.SP.6	Analyze whether a game is mathematically fair or unfair by explaining the probability of all possible outcomes.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
6.SP.7	Solve or identify solutions to problems involving possible combinations (e.g., if ice cream sundaes come in 3 flavors with 2 possible toppings, how many different sundaes can be made using only one flavor of ice cream with one topping?)	Grade 7, Mission 8, Topic B

7th Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
Ratios & Proportional Relationships		
7.RP.1	Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units.	Mission 2, Topic C Mission 4, Topic A
7.RP.2	Recognize and represent proportional relationships between quantities. Make basic inferences or logical predictions from proportional relationships. a. Decide whether two quantities are in a proportional relationship (e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin). b. Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships in real world situations. c. Represent proportional relationships by equations and multiple representations such as tables, graphs, diagrams, sequences, and contextual situations. d. Understand the concept of unit rate and show it on a coordinate plane. Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points $(0, 0)$ and $(1, r)$ where r is the unit rate.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Mission 3, Topic A Mission 3, Topic B Mission 4, Topic A Mission 5, Topic C Mission 5, Topic D
7.RP.3	Use proportional relationships to solve multistep ratio and percent problems.	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
The Number System		
7.NS.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram. a. Show that a number and its opposite have a sum of 0 (additive inverses). Describe situations in which opposite quantities combine to make 0. b. Understand addition of rational	Mission 5, Topic A Mission 5, Topic B Mission 6, Topic D Mission 7, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
	numbers ($p + q$ as the number located a distance q from p, in the positive or negative direction depending on whether q is positive or negative). Interpret sums of rational numbers by describing real-world contexts. c. Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in realworld contexts. d. Apply properties of operations as strategies to add and subtract rational numbers.	
7.NS.2	Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers and use equivalent representations. a. Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts. b. Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. If p and q are integers, then $-(p/q) = (-p)/q = p/(-q)$. Interpret quotients of rational numbers by describing real-world contexts. c. Apply and name properties of operations used as strategies to multiply and divide rational numbers. d. Convert a rational number to a decimal using long division; know that the decimal form of a rational number terminates in 0s or eventually repeats. e. Convert between equivalent fractions, decimals, or percents.	Mission 4, Topic A Mission 5, Topic A Mission 5, Topic C Mission 8, Topic A Mission 8, Topic B Mission 8, Topic D
7.NS.3	Solve real-world and mathematical problems involving the four operations with rational numbers. (Computations with rational numbers extend the rules for manipulating fractions to complex fractions.)	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Expression & Equations		

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
7.EE.1	Apply properties of operations as strategies to add, subtract, factor, expand and simplify linear expressions with rational coefficients.	Mission 6, Topic D
7.EE.2	Understand that rewriting an expression in different forms in a problem context can shed light on the problem and how the quantities in it are related.	Mission 6, Topic B
7.EE.3	Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form and assess the reasonableness of answers using mental computation and estimation strategies. Use variables to represent quantities in a real-world or mathematical problem, and construct multi-step equations and inequalities to solve problems by reasoning about the quantities. a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p , q , and r are specific rational numbers. b. Solve word problems leading to inequalities of the form $px + q > r$ or $px + q < r$, where p , q , and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem.	Mission 3, Topic C Mission 5, Topic C Mission 5, Topic F Mission 6, Topic A Mission 6, Topic B
7.EE.4	Use variables to represent quantities in a real-world or mathematical problem, and construct multi-step equations and inequalities to solve problems by reasoning about the quantities. a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p , q , and r are specific rational numbers. b. Solve word problems leading to inequalities of the form $px + q > r$ or $px + q < r$, where p , q , and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem.	Mission 5, Topic E Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic A
Geometry		
7.G.1	Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 2, Topic A Mission 3, Topic B Mission 3, Topic C

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
7.G.2	Draw (freehand, with ruler and protractor, and with technology) geometric shapes including polygons and circles with given conditions. Focus on constructing triangles from three measures of angles or sides, noticing when the conditions determine a unique triangle, more than one triangle, or no triangle.	Mission 3, Topic A Mission 7, Topic B Mission 7, Topic D
7.G.3	Describe the two-dimensional figures, i.e., cross-section, that result from slicing three-dimensional figures, as in plane sections of right rectangular prisms and right rectangular pyramids.	Mission 7, Topic C
7.G.4	Know the formulas for the area and circumference of a circle and use them to solve problems; give an informal derivation of the relationship between the circumference and area of a circle.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.G.5	Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle in a figure.	Mission 7, Topic A
7.G.6	Solve real-world and mathematical problems involving area, volume and surface area of two- and three-dimensional objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms.	Mission 1, Topic A Mission 2, Topic C Mission 3, Topic B Mission 7, Topic C Mission 7, Topic D
Statistics & Probability		
7.SP.1	Understand that statistics can be used to gain information about a population by examining a reasonably sized sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling tends to produce representative samples and support valid inferences.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
7.SP.2	Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
	samples) of the same size to gauge the variation in estimates or predictions.	
7.SP.3	Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, measuring the difference between the centers by expressing it as a multiple of a measure of variability.	Mission 8, Topic C Mission 8, Topic D
7.SP.4	Use measures of center and measures of variability for numerical data from random samples to draw informal comparative inferences about two populations.	Mission 8, Topic D Mission 8, Topic E
7.SP.5	Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around $\frac{1}{2}$ indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.	Mission 8, Topic A
7.SP.6	Approximate the probability of a chance event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability.	Mission 8, Topic A
7.SP.7	Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy. a. Design a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. b. Design a probability model (which may not be uniform) by observing frequencies in data generated from a chance process.	Mission 8, Topic A Mission 8, Topic C Mission 8, Topic E
7.SP.8	Find probabilities of compound events using organized lists, tables, tree diagrams, and simulation. a. Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which the compound event occurs. b. Represent sample spaces	Mission 8, Topic A Mission 8, Topic B

ALASKA MATHEMATICS STANDARDS	ZEARN MATH
7th Grade Standards	Lessons
for compound events using methods such as organized lists, tables and tree diagrams. For an event described in everyday language (e.g., “rolling double sixes”), identify the outcomes in the sample space which compose the event. c. Design and use a simulation to generate frequencies for compound events.	

8th Grade

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
The Number System		
8.NS.1	Classify real numbers as either rational (the ratio of two integers, a terminating decimal number, or a repeating decimal number) or irrational	Mission 8, Topic E
8.NS.2	Order real numbers, using approximations of irrational numbers, locating them on a number line.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
8.NS.3	Identify or write the prime factorization of a number using exponents.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Expressions & Equations		
8.EE.1	Apply the properties (product, quotient, power, zero, negative exponents, and rational exponents) of integer exponents to generate equivalent numerical expressions.	Mission 7, Topic B Mission 7, Topic C
8.EE.2	Use square root and cube root symbols to represent solutions to equations of the form $x^2 = p$ and $x^3 = p$, where p is a positive rational number. Evaluate square roots of small perfect squares and cube roots of small perfect cubes. Know that $\sqrt{2}$ is irrational.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
8.EE.3	Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities, and to express how many times as much one is than the other.	Mission 7, Topic C Mission 7, Topic D
8.EE.4	Perform operations with numbers expressed in scientific notation, including problems where both standard notation and scientific notation are used. Use scientific notation and choose units of appropriate size for measurements of very large or	Mission 7, Topic C Mission 7, Topic D

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
	very small quantities. Interpret scientific notation that has been generated by technology.	
8.EE.5	Graph linear equations such as $y = mx + b$, interpreting m as the slope or rate of change of the graph and b as the yintercept or starting value. Compare two different proportional relationships represented in different ways.	Mission 3, Topic A Mission 3, Topic B
8.EE.6	Use similar triangles to explain why the slope m is the same between any two distinct points on a non-vertical line in the coordinate plane; derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at b .	Mission 2, Topic C Mission 3, Topic B Mission 3, Topic C Mission 3, Topic E
8.EE.7	Solve linear equations in one variable. a. Give examples of linear equations in one variable with one solution, infinitely many solutions, or no solutions. Show which of these possibilities is the case by successively transforming the given equation into simpler forms, until an equivalent equation of the form $x = a$, $a = a$, or $a = b$ results (where a and b are different numbers). b. Solve linear equations with rational coefficients, including equations whose solutions require expanding expressions using the distributive property and combining like terms.	Mission 4, Topic B
8.EE.8	Analyze and solve systems of linear equations. a. Show that the solution to a system of two linear equations in two variables is the intersection of the graphs of those equations because points of intersection satisfy both equations simultaneously. b. Solve systems of two linear equations in two variables and estimate solutions by graphing the equations. Simple cases may be done by inspection.	Mission 3, Topic D Mission 3, Topic E Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
Geometry		
8.G.1	Through experimentation, verify the properties of rotations, reflections, and translations (transformations) to figures on a coordinate plane). a. Lines are taken to lines, and line segments to line segments of the same length. b. Angles are taken to	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 3, Topic B

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
	angles of the same measure. c. Parallel lines are taken to parallel lines.	
8.G.2	Demonstrate understanding of congruence by applying a sequence of translations, reflections, and rotations on two-dimensional figures. Given two congruent figures, describe a sequence that exhibits the congruence between them.	Mission 1, Topic C Mission 1, Topic D Mission 2, Topic B
8.G.3	Describe the effect of dilations, translations, rotations, and reflections on two-dimensional figures using coordinates	Mission 1, Topic A Mission 2, Topic A Mission 2, Topic C
8.G.4	Demonstrate understanding of similarity, by applying a sequence of translations, reflections, rotations, and dilations on two-dimensional figures. Describe a sequence that exhibits the similarity between them.	Mission 2, Topic B
8.G.5	Justify using informal arguments to establish facts about: the angle sum of triangles (sum of the interior angles of a triangle is 180°), measures of exterior angles of triangles, angles created when parallel lines are cut by a transversal (e.g., alternate interior angles), and angle-angle criterion for similarity of triangles.	Mission 1, Topic D Mission 2, Topic B Mission 2, Topic D
8.G.6	Explain the Pythagorean Theorem and its converse.	Mission 8, Topic C
8.G.7	Apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and mathematical problems in two and three dimensions.	Mission 8, Topic C
8.G.8	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.	Mission 8, Topic C
8.G.9	Identify and apply the formulas for the volumes of cones, cylinders, and spheres and use them to solve real-world and mathematical problems.	Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Statistics & Probability		

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
8.SP.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.	Mission 6, Topic A Mission 6, Topic B
8.SP.2	Explain why straight lines are widely used to model relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line, and informally assess the model fit by judging the closeness of the data points to the line.	Mission 6, Topic B
8.SP.3	Use the equation of a linear model to solve problems in the context of bivariate measurement data, interpreting the slope and y-intercept.	Mission 6, Topic B
8.SP.4	. Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects and use relative frequencies to describe possible association between the two variables.	Mission 6, Topic C
Functions		
8.F.1	Understand that a function is a rule that assigns to each input (the domain) exactly one output (the range). The graph of a function is the set of ordered pairs consisting of an input and the corresponding output. For example, use the vertical line test to determine functions and non-functions.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic E
8.F.2	Compare properties of two functions, each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).	Mission 5, Topic B Mission 5, Topic C
8.F.3	Interpret the equation $y = mx + b$ as defining a linear function, whose graph is a straight line; give examples of functions that are not linear.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E

ALASKA MATHEMATICS STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
8.F.4	Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a relationship or from two (x, y) values, including reading these from a table or from a graph. Interpret the rate of change and initial value of a linear function in terms of the situation it models, and in terms of its graph or a table of values.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
8.F.5	Given a verbal description between two quantities, sketch a graph. Conversely, given a graph, describe a possible real-world example.	Mission 5, Topic B Mission 5, Topic C